

The empirical recurrence for OEIS A299217, recovered from its tail

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Abstract

OEIS A299217 records “Empirical recurrence of order 66” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 72$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 3$, so the recurrence is proved for every $n \geq 69$. Its last coefficient is $c_{66} = -110592 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A299217 is “Number of $n \times 4$ 0..1 arrays with every element equal to 0, 2, 3, 4, 6 or 7 king-move adjacent elements, with upper left element zero.”. It has offset 1 and begins

1, 37, 81, 427, 2338, 13458, 84948, 543741, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Feb 05 2018 12:47:30, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 72$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 72$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 3$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 144$ exact terms from $n = 3$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	5	c_{18}	3997158	c_{35}	−1473188988	c_{52}	1308327266
c_2	20	c_{19}	−7670791	c_{36}	−911796652	c_{53}	1099481789
c_3	−10	c_{20}	−12906497	c_{37}	−574866398	c_{54}	769927004
c_4	−307	c_{21}	16659428	c_{38}	−818997657	c_{55}	327955104
c_5	−556	c_{22}	39815872	c_{39}	−223892745	c_{56}	−103145674
c_6	1075	c_{23}	12478237	c_{40}	1858193335	c_{57}	−277155736
c_7	4998	c_{24}	−31802582	c_{41}	3072665407	c_{58}	−128229134
c_8	6304	c_{25}	−29360827	c_{42}	3134168363	c_{59}	−30635380
c_9	−10341	c_{26}	12293717	c_{43}	3762034654	c_{60}	−15874828
c_{10}	−43945	c_{27}	39329758	c_{44}	4534202682	c_{61}	4314512
c_{11}	−17415	c_{28}	−137811894	c_{45}	5894440755	c_{62}	−1542360
c_{12}	140950	c_{29}	−248806252	c_{46}	5610943907	c_{63}	−2821104
c_{13}	81946	c_{30}	−217373519	c_{47}	3543629912	c_{64}	1684352
c_{14}	−539374	c_{31}	−482104346	c_{48}	1790056777	c_{65}	615168
c_{15}	−714065	c_{32}	−699908909	c_{49}	1030062084	c_{66}	−110592
c_{16}	370059	c_{33}	−986129500	c_{50}	853593001		
c_{17}	3179355	c_{34}	−1119487058	c_{51}	1163047277		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 69$, and it is the recurrence OEIS A299217 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 3$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{65} - \dots - c_{66}$. Its constant term is $-c_{66} = 110592$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$ vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the

minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 3)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 20 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -110592 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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